Aditya Parkhe

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TECHNICAL SKILLS

NLP & Deep Learning: PyTorch, HuggingFace Transformers, Fine-Tuning (LoRA/QLoRA), Tokenization, BLEU, ROUGE, BERT, Llama, SpaCy
Backend & Data: Python, FastAPI, ChromaDB, LangChain, PostgreSQL, Pandas, NumPy, Git, Docker, Linux
Techniques: Semantic Search, Grammar Correction, Style Transfer, Sequence-to-Sequence Modeling, Vector Embeddings

PROFESSIONAL EXPERIENCE

AI Engineer **Jan 2025 – Mar 2025**
Eipi Media *Mumbai, India*

- Evaluated and deployed state-of-the-art NLP models for automated text transformation and content analysis.
- Implemented prompt engineering heuristics and evaluation metrics, improving semantic coherence and tone consistency by 25%.
- Built automated data ingestion and text preprocessing pipelines, cutting raw text normalization time by 50%.

KEY PROJECTS

Transformer-Based Academic Grammar & Style Correction Engine github.com/adiP66/academic-nlp-assistant

- Fine-tuned sequence-to-sequence transformer models on academic editing corpora to perform automated grammar, syntax, and tone correction.
- Implemented token-level classification heuristics to highlight passive voice, stylistic redundancy, and journal guideline compliance.
- Achieved measurable gains on benchmark GLEU and edit-distance metrics, deploying lightweight quantized models via FastAPI inference endpoints.

Academic Research Literature RAG Assistant predmma.in

- Developed a dense vector search and citation attribution system using LangChain, ChromaDB, and OpenAI embedding models.
- Engineered semantic chunking pipelines that accurately preserve section context, formula tags, and bibliographic references.
- Implemented confidence score thresholds and cross-encoder re-ranking, mitigating hallucination rates below 2%.

Empirical Machine Learning Research Project github.com/adiP66/ufc-research

- Co-authored and publishing research investigating feature interpretability and model profitability in predictive sports analytics.
- Extracted and structured unstructured multi-source text and match reports using custom Python scrapers and regular expression engines.

EDUCATION

MSc. in Artificial Intelligence **2024 – 2026**
B.K. Birla College of Arts, Science & Commerce (University of Mumbai) *Kalyan, India*

BSc. in Computer Science **2021 – 2024**
Chandibai Himathmal Mansukhani College (University of Mumbai) *Ulhasnagar, India*

ACHIEVEMENTS

- Top 10% Most Active Contributor on LeetCode; solved 150+ algorithms and SQL problems.
- Authored comprehensive empirical research paper addressing model interpretability in predictive ML systems.